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Specific microbial ratio in the gut microbiome is associated with multiple sclerosis

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Gut microbiota dysbiosis is associated with multiple sclerosis (MS), but the causal relationship between specific gut bacteria and MS pathogenesis remains poorly understood. Therefore, we profiled the stool microbiome of people with MS (PwMS) and healthy controls (HC) using shotgun metagenomic sequencing. PwMS showed a distinct microbiome compared to HC, with *Prevotella copri* (*PC*) and *Blautia* species as drivers of microbial communities in HC and PwMS, respectively. Administration of MS-driving *Blautia* species (*Blautia wexlerae; BW*) to mice resulted in increased levels of gut inflammatory markers and altered microbiota with increased capacity to induce proinflammatory cytokines. Utilizing experimental autoimmune encephalomyelitis (EAE), an animal model of MS, we identified a lower gut *Bifidobacterium* to *Akkermansia* ratio as a hallmark of the disease. *BW*-administered mice also showed a lower *Bifidobacterium* to *Akkermansia* ratio pre-EAE induction which correlated with increased disease severity post-EAE induction. The importance of the *Bifidobacterium* to *Akkermansia* ratio at the species level, lower *Bifidobacterium adolescentis* to *Akkermansia muciniphila* (*BA:AM*), was validated in our MS cohort and a large International Multiple Sclerosis Microbiome Study. Thus, our findings highlight the *BA:AM* ratio as a potential gut microbial marker in PwMS, opening avenues for microbiome-based diagnosis, prognosis, and therapy in MS.

multiple sclerosis | gut microbiome | inflammation | microbial ratio | gut microbial markers

Multiple sclerosis (MS) is an inflammatory and demyelinating disease of the central nervous system (CNS) (1) with both genetic and environmental factors linked with MS predisposition. However, only 30% incidence in monozygotic twins highlights the significant influence of environmental factors in MS susceptibility (2). Recently, gut microbiota has emerged as an important environmental factor in the pathobiology of MS. The lack of microbiota in germ-free mice is linked with abnormal immune system (3, 4) and modifications in gut-motor functions, including delays in gastric emptying and intestinal transit (5, 6). Remarkably, nearly two-thirds of people with MS (PwMS) present with at least one or more impairments in gastrointestinal function (7). Recent studies have consistently linked gut microbial dysbiosis with PwMS and its animal model (8–15), highlighting the gut microbiome's role in MS pathogenesis and progression.

While previous studies have shown gut microbiome dysbiosis in PwMS (9, 10, 12, 14–18) and experimental autoimmune encephalomyelitis (EAE), an animal model of MS (8, 19), there is limited information on causality and no consensus on key microbial markers. Identifying key gut microbial markers associated with PwMS and validating their significance in animal models and human cohorts will help harness the enormous potential of the gut microbiota to develop diagnostic, prognostic, and therapeutic interventions.

Utilizing shotgun metagenomic sequencing of stool from PwMS, we identified *Prevotella copri* (*PC*) and *Blautia* species as the major drive of microbial community in healthy controls (HCs) and MS, respectively. Utilizing B6-MOG_{35-55}-EAE model, we observed a reduced ratio of *Bifidobacterium* to *Akkermansia* in feces and colonic mucosa as a characteristic feature of the gut microbiome dysbiosis. Administering MS-associated *Blautia* species (*Blautia wexlerae*; *BW*) to mice led to a lower *Bifidobacterium* to *Akkermansia* ratio in the gut and increased gut inflammatory markers pre-EAE. This lower ratio also correlated with heightened disease severity. Increased disease severity correlated with higher IL6 and IL-12/23 production by macrophages exposed to fecal lipopolysaccharides (LPS) from mice treated with MS-associated *BW*. A lower fecal *Bifidobacterium* to *Akkermansia* ratio at the species level; *Bifidobacterium adolescentis* to *Akkermansia muciniphila* (*BA:AM*) was observed in our MS cohort and validated in a larger cohort study [International

Significance

We demonstrate the heterogeneity of the gut microbiome in multiple sclerosis (MS) patients, characterized by the enrichment of *Blautia* and *Akkermansia* species. Using a mouse model of MS, we identified that a reduced *Bifidobacterium* to *Akkermansia* ratio served as a gut microbial marker of disease severity. Administering MS-associated *Blautia* species in mice altered this ratio and strongly correlated with disease scores, underscoring its potential in predicting severity. The lower *Bifidobacterium* to *Akkermansia* ratio at the species level, *Bifidobacterium adolescentis* to *Akkermansia muciniphila* (*BA:AM*), was validated in our human cohort and the larger International Multiple Sclerosis Microbiome Study (iMSMS). These findings highlight the *BA:AM* ratio's promise as a microbial marker for microbiome-based diagnostics and therapeutic interventions in MS.

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Table 1. Demographics and clinical characteristics of HCs and RRMS patients who provided stool samples for the metagenomic study

Criteria	MS (n = 45)	HCs (n = 51)	*P*-value
Age (mean ± SD)	41.8 ± 8.87	40.71 ± 14.69	*P* = 0.65
Age (<40 y: >40 y)	20:25	26:25	*P* = 0.54
Sex (Male:Female)	11:34	8:43	*P* = 0.31
Treatment (Yes:No)	35:10	0:51	
Tecfidera	14	–	–
Copaxone	5	–	–
Avonex	3	–	–
Tysabri	1	–	–
Ocrelizumab/Ocrevus	11	–	–
Rebif	1	–	–

Data are provided as n, mean (SD), or range. Age: Unpaired *t* test with Welch's correction, Age (<40 y:>40 y): Fisher's exact test, Sex: Fisher's exact test.

Multiple Sclerosis Microbiome Study (iMSMS)]. A lower *BA:AM* ratio was associated with higher Expanded Disability Status Scale (EDSS) scores, emphasizing its role as an important gut microbial determinant in PwMS. Our findings provide insight into mechanism of gut dysbiosis in MS and may inform future microbiome-based diagnostics and therapies.

Results

Cohort Characteristics. Forty-five patients with relapsing–remitting MS (RRMS) and 51 age- and sex-matched HCs were recruited for this study (Table 1). Our MS cohort exhibited a nearly 3:1 female-to-male ratio which was not different from the HC cohort. At the time of sample collection, no patients had active disease, 35 PwMS were on disease-modifying treatments (DMTs) and 10 were untreated. However, there was no significant difference between the ratio of participants below or above 40 y of age; or in the ratio between males and females.

PwMS Exhibit Inflammatory Fecal and Blood Biomarkers. Given the frequent gastrointestinal dysfunction in PwMS (20, 21), we investigated gut inflammation by measuring fecal lipocalin2 and calprotectin levels. While fecal lipocalin2 levels were significantly elevated in PwMS compared to HCs (P = 3e-04) (Fig. 1*A*), fecal calprotectin levels were not (P = 0.9) (Fig. 1*B*). Plasma osteopontin, a systemic circulatory biomarker linked with MS (22), was significantly elevated in PwMS (P = 9.6e-06) (Fig. 1*C*). Both treated and untreated PwMS showed higher fecal lipocalin-2 and plasma osteopontin levels compared to HCs, with no difference between the two MS groups (*SI Appendix*, Fig. S1*A*). Fecal calprotectin levels were similar between HCs and PwMS, regardless of treatment status (*SI Appendix*, Fig. S1*A*). Furthermore, fecal lipocalin2, fecal calprotectin, and plasma osteopontin levels were all positively correlated (*SI Appendix*, Fig. S1*B*). These data suggest a link between intestinal and systemic inflammation, potentially driven by gut microbiome dysbiosis in PwMS.

PwMS Harbor *Blautia*- and *Akkermansia*-Driven Taxonomically and Functionally Distinct Gut Microbiome. Fecal microbiome profiling using shotgun metagenomic sequencing revealed no significant differences in Shannon or phylogenetic diversity between HCs and PwMS, regardless of the treatment status (Fig. 1 *D* and *E* and *SI Appendix*, Fig. S1 *C* and *D*). Similarly, no differences in Shannon or phylogenetic diversity were observed when stratifying by age (<40 y, >40 y), sex (male, female), and body mass index (BMI; normal, obese, overweight) (*SI Appendix*, Fig. S1 *E* and *F*). However, there were distinct differences in the β-diversity between HCs and PwMS (R^2 = 0.027, P = 0.001) (Fig. 1*F*). Significant differences in microbiome composition between DMT-treated PwMS (R^2 = 0.026, P = 0.018) or untreated PwMS (R^2 = 0.032, P = 0.022) compared to HCs but not between DMT-treated and untreated PwMS groups (R^2 = 0.023, P = 0.39) (*SI Appendix*, Fig. S1*G*), indicated that treatment did not affect the overall gut microbiome composition in PwMS. Gut microbiome community compositions differed significantly between HCs and PwMS when stratified by age (>40 y: R^2 = 0.048, P = 0.005) (*SI Appendix*, Fig. S2*A*), sex (female: R^2 = 0.032, P = 0.002; male: R^2 = 0.1, P = 0.02) (*SI Appendix*, Fig. S2*B*), BMI (normal, R^2 = 0.042, P = 0.01; obese, R^2 = 0.068, P = 0.031) (*SI Appendix*, Fig. S2*C*). However, no differences were observed in the microbiome composition between HCs and PwMS with those <40 y old (R^2 = 0.029, P = 0.12) or BMI categorized as "overweight" (R^2 = 0.045, P = 0.31) (*SI Appendix*, Fig. S2 *A* and *C*). Notably, PwMS displayed distinct gut microbiome compositions compared to HCs across both normal and obese BMI categories (*SI Appendix*, Fig. S2*C*), suggesting that MS exerts a stronger influence on gut microbiota than BMI alone. Overall, our data suggest a distinct gut microbiome composition in PwMS compared to HC.

As treatment had no effect on gut microbiome composition, we included all MS samples for taxonomic analysis. There were no differences in the relative abundance taxa at the phylum level (*SI Appendix*, Fig. S3*A*) and the *Bacteroidetes* to *Firmicutes* ratio between HCs and PwMS (*SI Appendix*, Fig. S3*B*). By overlaying species abundance onto a Bray–Curtis–based PCoA plot of HCs and PwMS, we identified *PC* in HC and *Blautia species* (Fig. 2*A*) in PwMS as the key bacterial drivers of microbiome community. Moreover, *Blautia* species were also key drivers in untreated MS patients, and *Phocaeicola vulgatus* was for DMT-treated PwMS (*SI Appendix*, Fig. S4). Upon differential analysis, *PC, Faecalibacterium prausnitzii*, *B. adolescentis, etc.,* were differentially abundant in HCs compared to PwMS (Fig. 2*B* and Dataset S1). Conversely, PwMS showed enrichment of *Collinsella, Eggerthella, Dorea* asp: CAG:317, and numerous *Ruminococcus* species including *Blautia* sp. CAG:257 (Fig. 2*B*). Thus, *Blautia*/*Ruminococcus* species and *Prevotella* species might be key drivers of MS and HC microbiome communities, respectively.

To address reported gut microbiome differences in MS observed in this and previous studies (9, 12, 14–16, 18) and to identify whether *Blautia* species are key to PwMS gut microbiota, we used *topic modeling,* a statistical modeling approach with an unsupervised machine learning method on our metagenomic sequencing data. Topic modeling has been used for identifying bacterial communities in microbiome data relevant to health and disease state where each topic corresponds to a potential bacterial community (23, 24). Using this approach, we identified 39 different topics in our cohort (the lowest average of "CaoJuan2009" and "Arun2010" metrics) (*SI Appendix*, Fig. S5*A*). No specific microbial topics were enriched in HCs compared to PwMS (*SI Appendix*, Fig. S5*B*). However, four distinct topics were enriched in PwMS: specifically, topics 1, 8, 23, and 35 (*SI Appendix*, Fig. S5 *C–F*). *Blautia*/ *Ruminococcus* and *Akkermansia* species exhibited a higher probability of being assigned to topics 1, 8, 23, and 35, respectively, enriched in PwMS communities. *Blautia* species were assigned to three MS topics (1, 8, and 35), highlighting their importance in the gut microbiome of PwMS. Overall, our findings suggest that multiple microbial species may be associated with PwMS communities, explaining inconsistencies across previous

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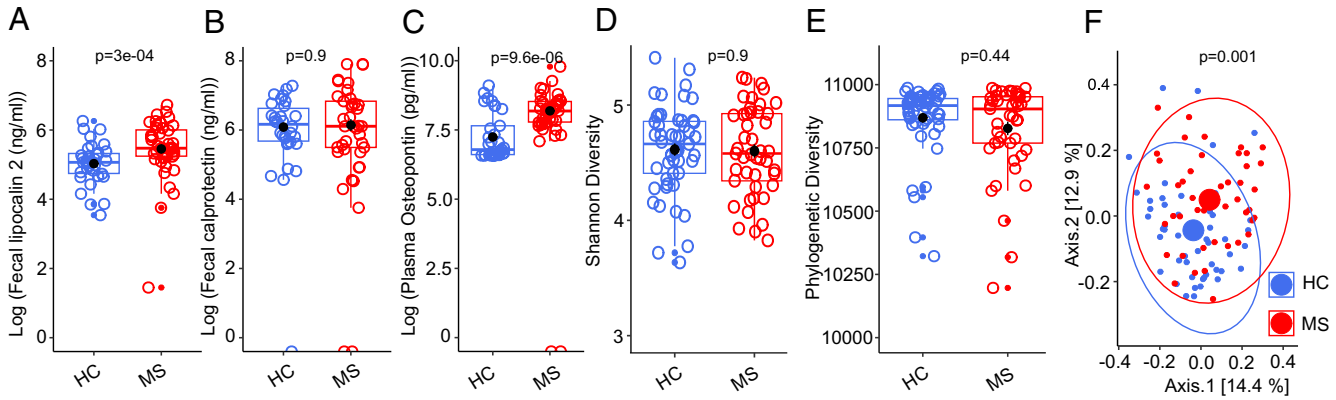


Fig. 1. PwMS (people with MS) harbor distinct inflammatory gut microbiome irrespective of treatment. (*A*–*C*) Quantification of (*A*) fecal lipocalin2, (*B*) fecal calprotectin, and (*C*) plasma osteopontin levels in samples from HCs and PwMS using the ELISA. For fecal lipocalin2 and calprotectin, samples from 34 HC and 44 PwMS were included in the analysis as others were beyond detection range in the ELISA. For plasma osteopontin, samples from 33 HCs and 44 PwMS were included in the analysis as others were beyond the detection range or did not provide associated blood samples. (*D*) Shannon diversity between and (*E*) phylogenetic diversity between PwMS and HCs. (*F*) Bray–Curtis dissimilarity metric–based principal coordinate analysis plot depicting compositional differences in the microbial communities of PwMS and HCs. *P*-values represent statistical significance using the Wilcoxon test in (*A*–*E*). *adonis2* test from the *vegan* package in R was used for significance testing in (*F*).

studies. Additionally, *Blautia* species appear to be common and influential drivers of the gut microbiome in PwMS.

Gut microbiota substantially affects metabolic, immunological, and protective functions, eventually affecting disease outcome (25–27). When analyzing functional capabilities, we found differences in the gut microbiome pathways between HCs and PwMS. Although no differences were observed in the overall functions of the PwMS and HCs gut microbiome ($P = 0.24$) (*SI Appendix*, Fig. S6*A*), there were 51 different KEGG Orthologs (KOs) between the PwMS and HCs (Dataset S2). To identify gut-derived metabolites potentially relevant to MS, differentially expressed KOs were linked to the gut brain modules (GBMs) and gut metabolic modules (GMMs). We found two matching to GBMs and nine matching to GMMs, respectively (Dataset S2). Among GBMs, phosphoriosylanthranilate isomerase (*trpF*) was elevated, while myo-inositol-1-phosphate synthase (*INO1*) was diminished in PwMS (*SI Appendix*, Fig. S6 *B* and *C*). Among in GMMs, mannose-6-phosphate isomerase (*manA*), ribose-5-phosphate isomerase (*rpiB*), L-rhamnose isomerase (*rhaA*), histidinol-phosphate aminotransferase (*hisC*), and aconitate hydratase (*acnA*) exhibited lower levels, while glucokinase (*glk*), alpha-L-fucosidase (*FUCA*), glucose-6-phosphate isomerase (*pgi*), and fructose-bisphosphate aldolase class II (*fbaA*) showed higher levels in PwMS (*SI Appendix*, Fig. S6 *D* and *L*). Distinct gut and brain modules in HCs and PwMS suggest differential effect of the gut microbiome on CNS, with *Blautia* and *Akkermansia* species driving taxonomic and functional differences in PwMS.

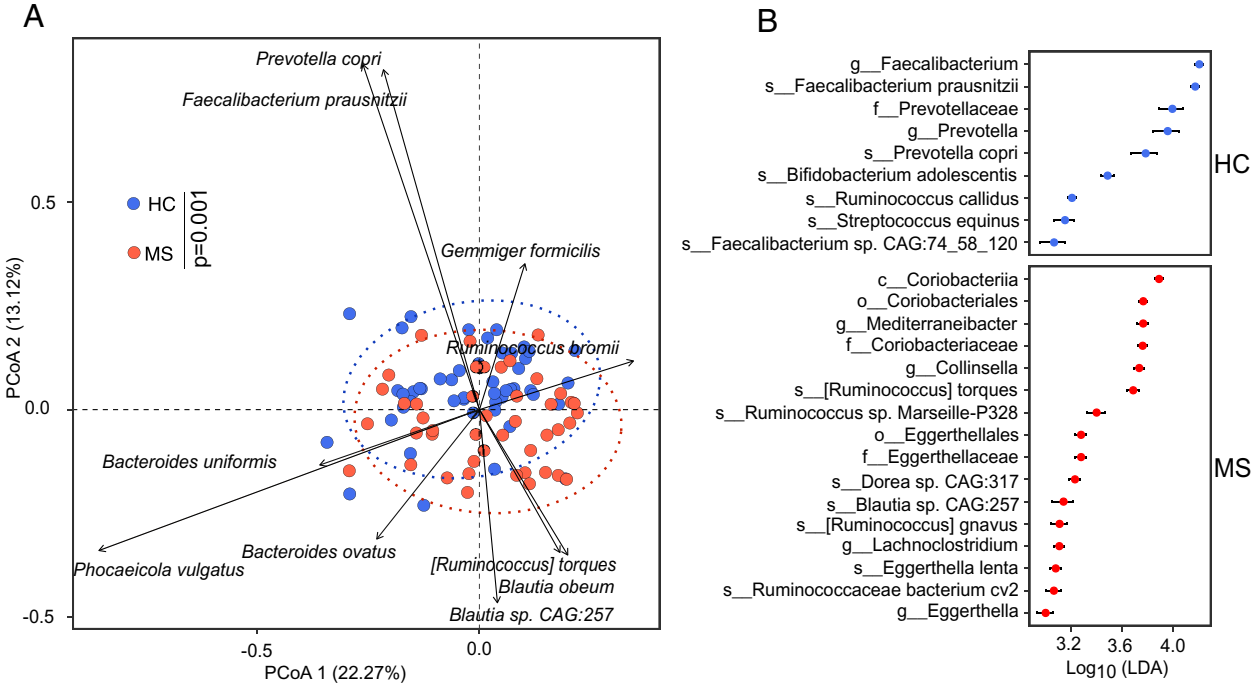


Fig. 2. *Blautia* and/or *Akkermansia* drive the heterogeneous gut microbiome in PwMS. (*A* and *B*) Bray–Curtis dissimilarity metrics–based PCoA plot representing major bacterial taxa weighing toward samples from HCs or PwMS. *adonis2* test from the *vegan* R package was used for statistical inference. (*B*) Differential analysis of the microbiome between the HCs and PwMS microbiome was performed using *diff_analysis* function from the *microbiotaprocess* package in R with other parameters being default except *lda* = 3 and $P < 0.05$ and default $q < 0.3$ in samples from HCs and PwMS and visualized using *ggplot2*.

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A Lower Ratio of *Bifidobacterium:Akkermansia* Is a Feature of an EAE Gut Microbiome. As the gut microbiome is heterogeneous in humans, we used EAE, a well-established mouse model of MS, to identify disease-related gut microbiota markers. C57BL/6J (B6) female mice (4 to 6 wk old) were assigned randomly into three groups (n = 5/group), as shown in Fig. 3*A*, where each group received either PBS (control), CFA+PTX (vehicle control), or MOG_{35-55}+CFA+PTX (EAE) as outlined in *Methods*. As expected, only the group administered MOG_{35-55}+CFA+PTX developed paralytic disease (Fig. 3*B*). Fecal microbiota analysis at baseline (day 0), predisease (day 7), and the peak of the disease (day 14) showed that although there was no difference in the Shannon diversity at early time points, the MOG_{35-55}-immunized group exhibited significantly lower Shannon diversity compared to the vehicle control on day 14 (Fig. 3*C*). While fecal microbiome composition was similar at baseline (Fig. 3*D*), significant differences emerged between groups by day 7 and become more pronounced by day 14 in mice with EAE compared to controls (Fig. 3*D*). *Akkermansia* species were enriched in MOG_{35-55}-immunized mice, while *Lactobacillus* and *Faecalibaculum* were enriched in the PBS control and *Bifidobacterium* species were enriched in the vehicle control group (Fig. 3*E* and *SI Appendix*, Fig. S7*A*). In colonic

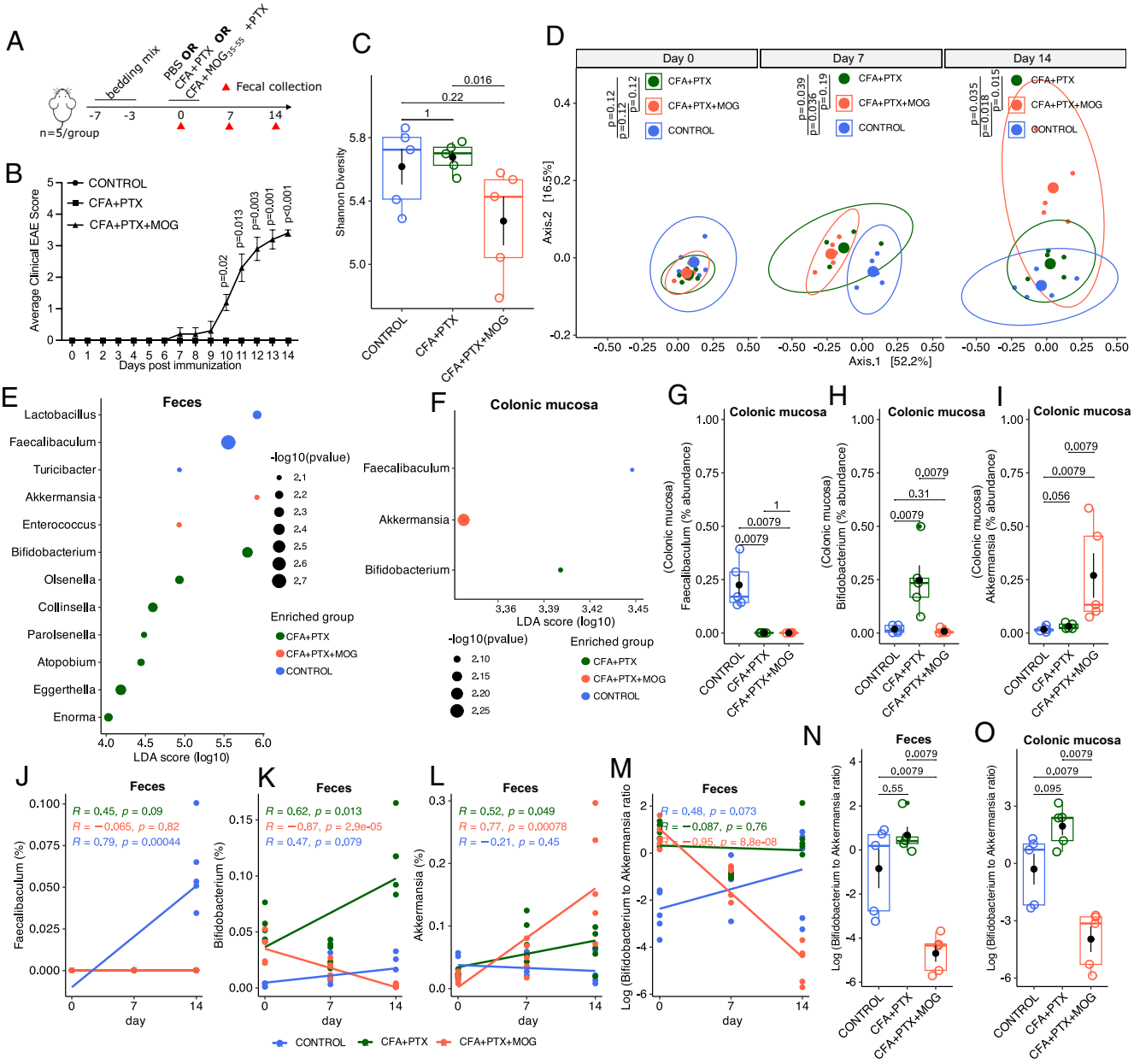


Fig. 3. Gut microbiota profile in mice induced with EAE. C57BL/6J female mice (4 to 6 wk old) were obtained from Jackson Laboratories (Bar Harbor, ME) and bred in the University of Iowa mouse facility. Pups at the age of 4 to 6 wk were kept on a standard normal chow diet ad libitum and randomly divided into three groups (n = 5/group). Mice in each group received PBS (control), CFA+PTX (vehicle control), or CFA+PTX+MOG_{35-55} (EAE). (*A*) Schematic diagram of the experiment. (*B*) Average clinical EAE scores (mean ± SEM). (*C*) Shannon diversity on day 14. (*D*) PCoA plots showing microbiome compositional differences between groups on days 0, 7, and 14 based on Bray–Curtis dissimilarity metrics. *P*-value determined by *adonis2* test followed by pairwise comparison and "BH" correction. Microbial taxa were collapsed at the genus level and differentially abundant taxa from (*E*) fecal samples and (*F*) colonic mucosa in the three groups on day 14, as determined using the "*lefse*" function in the "*microbiomeMarker*." (*G–I*) Relative abundance (%) of bacteria enriched in colonic mucosa for (*G*) *Faecalibaculum,* (*H*) *Bifidobacterium,* and (*I*) *Akkermansia* in figure (*F*) in each of the three groups. (*J–L*) Fecal *Faecalibaculum*, *Bifidobacterium,* and *Akkermansia* abundances in feces of three groups over time (*M*) *Bifidobacterium* to *Akkermansia* ratio in feces between the three groups. (*N*) *Bifidobacterium* to *Akkermansia* ratio in feces between the three groups on day 14. (*O*) *Bifidobacterium* to *Akkermansia* ratio in colonic mucosa between the three groups on day 14. The Kruskal–Wallis test followed by pairwise comparison and "BH"-based *P*-value correction was used for group-wise comparison for figures (*C*, *G–I*, *N*, and *O*).

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mucosa (day 14), only *Faecalibaculum, Bifidobacterium*, and *Akkermansia* species were significantly enriched across all groups (Fig. 3 *F* and *G–I* and *SI Appendix*, Fig. S7*B*). Relative abundances of fecal *Faecalibaculum, Bifidobacterium,* and *Akkermansia* altered distinctly over time with disease progression in PBS-treated, vehicle control, and MOG_{35-55}-immunized groups, respectively (Fig. 3 *J* and *L*). Interestingly, *Bifidobacterium* and *Akkermansia* followed opposite trends, with *Bifidobacterium* significantly increasing in the vehicle control and decreasing in the MOG_{35-55}-immunized group, and vice versa for *Akkermansia* (Fig. 3 *K* and *L*).

As the vehicle control serves as the true control for the MOG_{35-55}-immunized group and given opposing *Bifidobacterium*/*Akkermansia* changes in MOG_{35-55} vs. vehicle control, disease severity likely stems from altered host signaling due to the changes in relative abundance of *Akkermansia* and *Bifidobacterium* in the gut. Thus, to capture the changes in relative abundances of both bacteria, we examined the ratio of *Bifidobacterium* to *Akkermansia* change over time in feces and found that *Bifidobacterium* to *Akkermansia* ratio was significantly lower only in the MOG_{35-55}-immunized group and correlated with disease progression over time (Fig. 3*M*). Specifically, the ratio was significantly lower in fecal and colonic mucosal samples on peak disease (day 14) (Fig. 3 *N* and *O*). Thus, our data suggest that the *Bifidobacterium* and *Akkermansia* ratio can be a gut microbial feature in EAE disease.

Administering MS-Driving BW Results in Proinflammatory Gut with a Unique Gut Microbiome and a Lower *Bifidobacterium: Akkermansia* Ratio. While a lower *Bifidobacterium* to *Akkermansia* ratio correlated with EAE disease, we argued that this ratio can predict EAE severity. We hypothesize that the administration of MS-driving *Blautia* species will alter the gut microbiome to lower the *Bifidobacterium* to *Akkermansia* ratio, enhance gut inflammation, and subsequently increase EAE severity compared to the administration of HC-driving bacteria. Female B6 mice (4 to 6 wk old) were kept on a standard normal chow diet ad libitum and randomly divided into three groups (n = 9-10/group) followed by treatment with antibiotics cocktail to deplete the existing gut microbiome and overcome colonization resistance of the gut microbiome for the administered bacterium as described previously (28) and detailed in the *Methods* section. Following antibiotic treatment, mice were orally gavaged with MS-driving *Blautia* species (*B. wexlerae*; *BW*) or HC-driving *PC* as described (Fig. 4*A*). *PV*, a bacterium weighing toward MS-treated group (*SI Appendix*, Fig. S4) was cultured in-house from healthy human fecal samples as described in the *Methods* section and used as a control.

Administration of *PC, PV,* and *BW* altered the gut microbiome richness before EAE induction, on day 27 where the *PC*-administered group had significantly lower Shannon diversity in the microbiome than mice administered *PV* or *BW* (*SI Appendix*, Fig. S8*A*). *PC*-, *PV*-, and *BW*-administered groups followed a unique path for microbial community assembly over time (*SI Appendix*, Fig. S8*B*). There was a significant difference in the *β-diversity* among groups administered with *PC, PV,* or *BW* at days 13 (R^2 = 0.74, *P* = 9.99e-05), 17 (R^2 = 0.32, *P* = 9.99e-05), 22 (R^2 = 0.45, *P* = 9.99e-05), and 27 (R^2 = 0.63, *P* = 9.99e-05) (*SI Appendix*, Fig. S8*C*). On day 27, *PC*-administered mice had significantly higher abundances of *Bacteroidota* and lower levels of *Desulfobacterota*, *Proteobacteria*, and *Verrucomicrobiota* compared to the other two groups (*SI Appendix*, Fig. S8*D*). However, the *BW*-administered group had significantly higher levels of *Proteobacteria* and *Verrucomicrobiota,* whereas the *PV*-administered group had significantly higher levels of *Desulfobacterota*, *Actinobacteriota,* and *Firmicutes* (*SI Appendix*, Fig. S8*D*). At the genus level 26 taxa were differentially abundant across three groups (Fig. 4*B*). *Bacteroides, Clostridum_sensu_stricto_1, Turicibacter, Marvinbryantia, Enterococcus, Romboutsia,* and

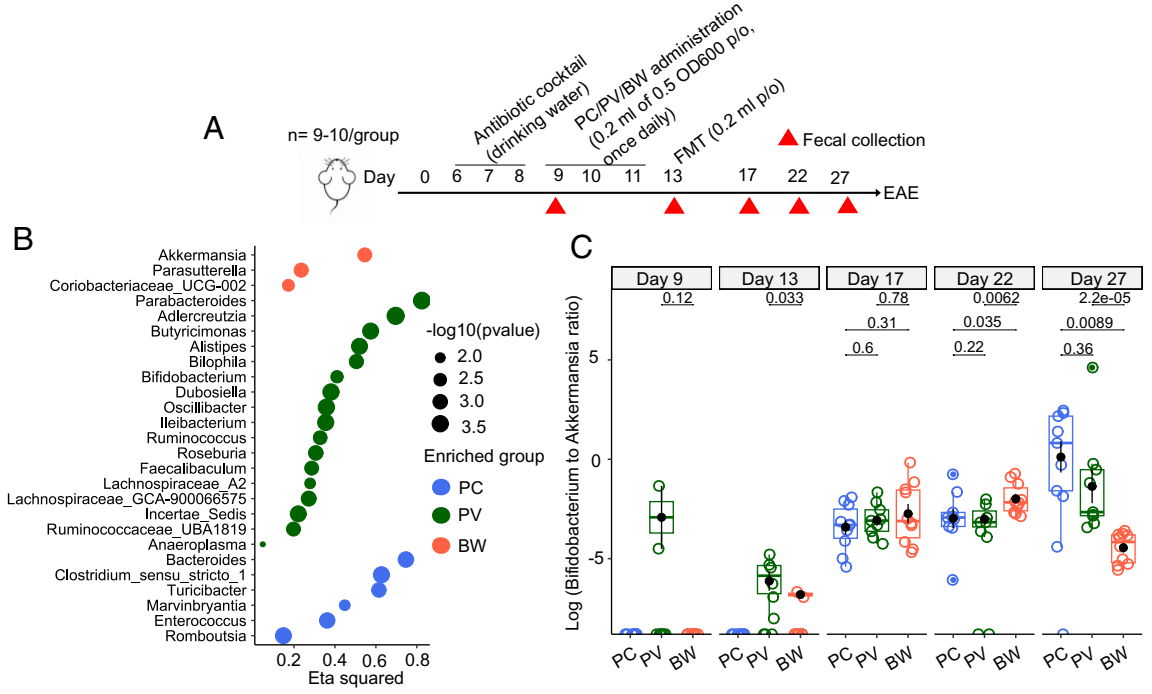


Fig. 4. Alteration of gut microbiome communities in *PC*-, *PV*-, and *BW*-administered mice. (*A*) Schematic experimental outline. C57BL/6J female mice (4 to 6 wk old) were randomly divided into three groups (n = 9-10/group) and treated with antibiotics cocktail (combination of vancomycin, neomycin, ampicillin, and metronidazole supplemented with Splenda) for a 3 d-period to deplete the gut microbiome. Mice in each group were then administered with *PC, PV,* or *BW* followed by FMT. Fecal samples were collected at different time points as shown by red triangles for microbiome community assessment. (*B*) Differentially enriched bacterial taxa at the genus level after mice were administered *PC, PV,* or *BW* (day 27). *P*-adjusted values ≤ 0.05 was used for significance testing. (*C*) *Bifidobacterium:Akkermansia* ratio over time on days 9, 13, 17, 22, and 27 in *PC*-, *PV*-, and *BW*-administered mice. The Kruskal–Wallis test followed by pairwise comparison and "BH"-based *P*-value correction was performed in (*C*).

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Merdibacter were enriched in the *PC*-administered group, whereas *Akkermansia, Parasutterella,* and *Coriobacteriaceae_UCG-002* were enriched in the *BW*-administered group. The *PV*-administered group showed enrichment of 17 additional genera including *Parabacteroides, Adlercreutzia,* and *Butyricimonas* were enriched in the (Fig. 4*B*). Knowing the importance of the *Bifidobacterium* to *Akkermansia* ratio in EAE disease severity, we analyzed their abundance and ratio across groups. *Bifidobacterium* was significantly higher in the *PV*-administered mice compared to both *PC*- and *BW*-administered mice (*SI Appendix*, Fig. S8*E*). In addition, *Akkermansia* was significantly higher in both *PV*- and *BW*-administered groups compared to *PC*-administered mice (*SI Appendix*, Fig. S8*F*). While there was no initial difference (day 9, 13 and 17) in the *Bifidobacterium* to *Akkermansia* ratio, by day 27, mice receiving MS-driving *BW* had significantly lower *Bifidobacterium* to *Akkermansia* ratio compared to both HC-driving *PC*- and *PV*-administered groups on day 27 (Fig. 4*C*). This suggests that introducing specific bacteria can have a "priority effect," influencing how the microbiome community develops.

As *PC, PV*, and *BW* administrations resulted in varying *Bifidobacterium* to *Akkermansia* ratios, we investigated whether altered ratio also affects inflammatory responses. Indeed *BW*-administered mice showed high fecal lipocalin2 and calprotectin levels on day 27 (Fig. 5 *A* and *B*), suggesting *BW*-administration caused both gut dysbiosis and higher gut inflammation. Furthermore, fecal lipopolysaccharide (LPS) isolated from MS-driving *BW*-administrated mice induced significantly higher levels of TNFα, IL1β, and IL-12/23 but not CXCL1 or IL-10 from bone marrow–derived macrophages (BMDMs) (Fig. 5 *C* and *G*). Thus, the administration of specific bacterium can significantly affect the priming of gut microbiota to become pro- or anti-inflammatory. Interestingly, we observed a striking negative relationship between the *Bifidobacterium* to *Akkermansia* ratio and the levels of fecal inflammatory markers (lipocalin2 and calprotectin) in *PC-, PV-*, or *BW*-administered mice on day 27 (Fig. 5 *H* and *I*). Additionally, the *Bifidobacterium* to *Akkermansia* ratio in *PC-, PV-*, or *BW*-administered mice on day 27 negatively correlated with the production of inflammatory cytokines TNFα, IL6, and IL-12/23 but not with CXCL1 or IL10 (Fig. 5 *J* and *N*). Thus, our data strongly indicated that the balance between *Bifidobacterium* and *Akkermansia* in the gut could be a key indicator of inflammatory activity.

Finally, we investigated whether the *PC-, PV-*, and *BW*-administered mice impacted EAE severity. Indeed, *BW*-administered mice developed severe EAE disease compared to *PC*- or *PV*-administered mice (Fig. 6*A*). However, we did not find a significant difference in the cumulative EAE scores between *PC-, PV-*, and *BW*-administered mice (Fig. 6*B*). In addition, the predisease (day 27) *Bifidobacterium* to *Akkermansia* ratio showed a strong negative correlation with EAE severity, highlighting the potential of this ratio as an early biomarker for disease severity (Fig. 6*C*). Although fecal lipocalin2 and calprotectin levels showed no correlation with the cumulative EAE scores (Fig. 6 *D* and *E*), there was a strong correlation between cumulative EAE scores and immunostimulatory ability of fecal LPS isolated from *PC-, PV-*, and *BW*-administered mice. Specifically, a positive correlation was observed between cumulative EAE scores (post-EAE) and fecal LPS-induced cytokines IL6 and IL-12/23 production from BMDMs (pre-EAE induction, day 27) (Fig. 6 *F* and *J*). Thus, our data indicate a potential causal link between gut dysbiosis and change in *Bifidobacterium* to *Akkermansia* ratio, characterized by immunostimulatory LPS, heightened gut inflammation, and EAE severity.

A Lower *BA:AM* Ratio Is a Gut Microbial Marker in PwMS. Next, we investigated whether a low *Bifidobacterium* to *Akkermansia* ratio could also serve as a potential biomarker of MS in our human cohort. Although overall *Bifidobacterium* to *Akkermansia* ratio at the genus level was not different between HCs and PwMS ($P = 0.35$), a deeper look at the species level in our cohort showed that *B. adolescentis (BA)* was differentially abundant in HC (Fig. 2*B*). Therefore, we compared the abundance of *BA* to all *Akkermansia*

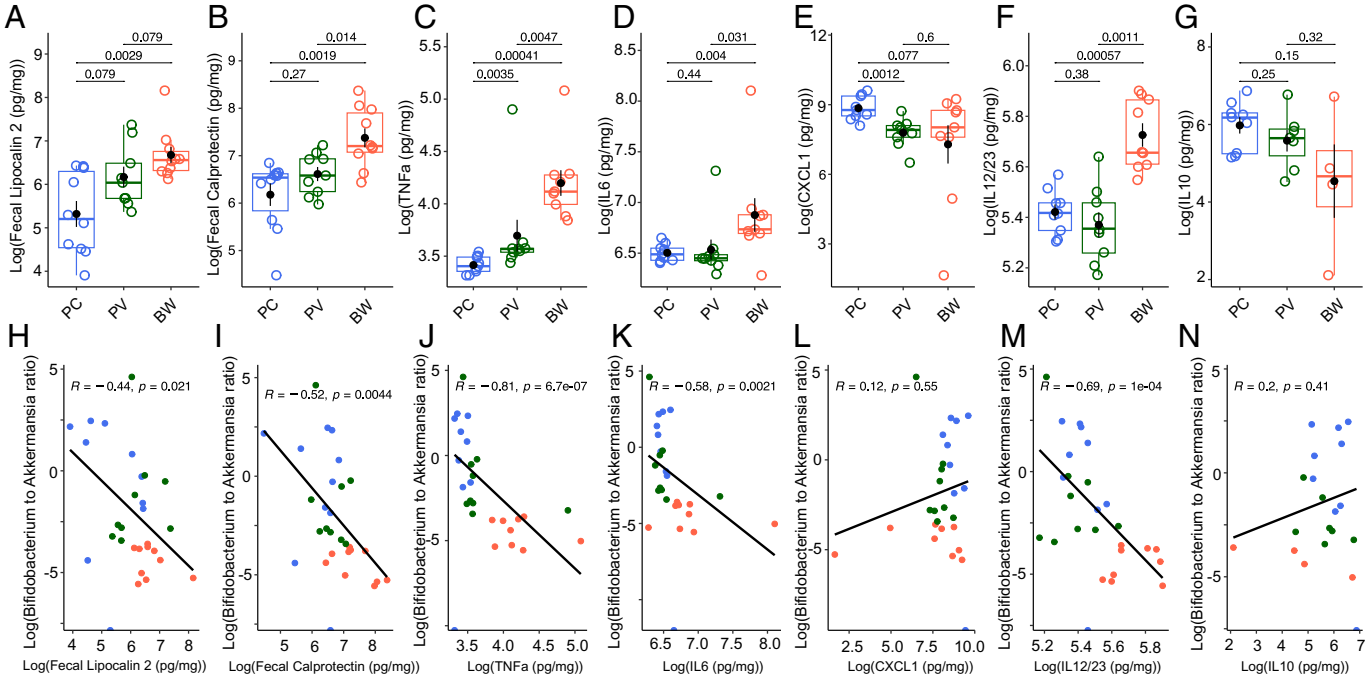


Fig. 5. Administering *Blautia* species increases gut inflammatory markers and lowers *Bifidobacterium:Akkermansia* ratio. (*A* and *B*) Quantification of fecal lipocalin2 and calprotectin on day 27. (*C–G*) Determination of levels of cytokines TNF α (*C*), IL6 (*D*), CXCL1 (*E*), IL12/23 (*F*), and IL10 (*G*) in supernatants of BMDMs cultured with fecal lipopolysaccharides extracted from fecal sample of day 27 mice were administered *PC*, *PV*, or *BW*. (*H–N*) Spearman correlation of *Bifidobacterium:Akkermansia* ratio to fecal lipocalin2, calprotectin, and cytokines levels in (*A–G*). The Kruskal–Wallis test followed by pairwise comparison and "BH"-based *P*-value correction was performed for (*A–G*) between the groups.

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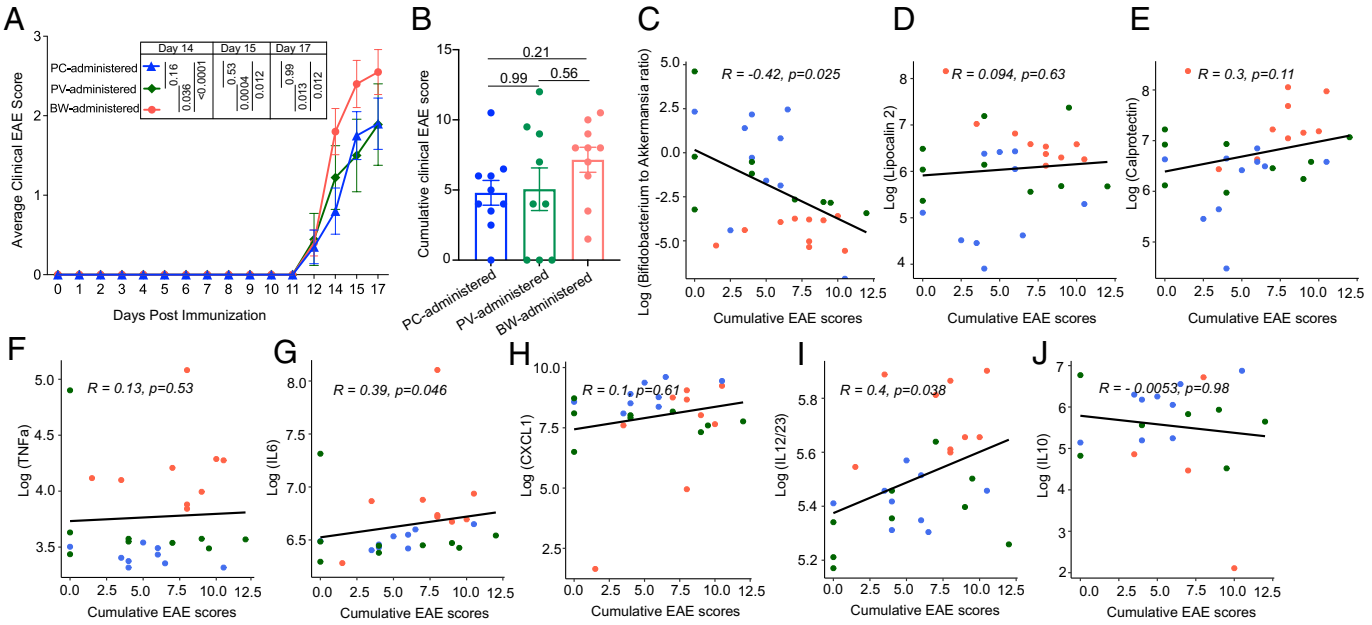


Fig. 6. Increased EAE severity in mice administered with *BW* is associated with lower *Bifidobacterium:Akkermansia* ratio and elevated IL6 and IL-12/23. (*A*) Average clinical EAE scores (mean ± SEM) of mice administered with *PC, PV,* and *BW* over time after day 27 in Fig. 4*A*. Two-way ANOVA was performed to compare mean EAE scores of mice in each group with other group on a specific day and *P*-values represent *P*-adjusted values that were obtained after correcting for multiple comparison using the Tukey method in GraphPadPrism10.3.0. No significance was observed between groups on day 0 to day 12. Significant differences with adjusted-*P*-values on day 14, 15 and 17 are shown in figure. (*B*) Cumulative EAE scores of mice from (*A*). One-way ANOVA using Brown–Forsythe and Welch ANOVA tests was used for statistical comparison in GraphPadPrism10.3.0. (*C–J*) Spearman correlation of pre-EAE induction (day 27) *Bifidobacterium:Akkermansia* ratio to (*C*) cumulative EAE scores (*D*) lipocalin2 (*E*) calprotectin (*F*) TNFα (*G*) IL6 (*H*) CXCL1 (*I*) IL-12/23 and (*J*) IL10 levels from Fig. 5 *A–G*.

species identified in our cohort [including *A. muciniphila (AM)*, *Akkermansia sp. CAG:344*, *Akkermansia sp. KLE1797*, and *A. glycaniphila*]. We observed that only the *BA*:*AM* ratio was significantly lower in PwMS compared to HC (P = 0.041) and abundance of *BA* but not *AM* was significant different between the HC and MS groups in our cohort (Fig. 7 *A–C*). The ratio of *BA* to other *Akkermansia* species was not different (*Akkermansia* sp. CAG:344, P = 0.066; *Akkermansia* sp. KLE1797, P = 0.097; and *A. glycaniphila*, P = 0.096). These results suggest that the *BA:AM* ratio may be an important marker of MS disease in humans.

The significance of the *BA:AM* ratio as a potential MS biomarker was further corroborated by analysis of a larger dataset (iMSMS) comprising 572 paired household controls (HC) and 572 individuals with MS (PwMS) (15). In iMSMS data, *BA* was higher in HC, whereas *AM* were higher in PwMS (Fig. 7 *D* and *E*) and the *BA:AM* ratio was significantly lower in PwMS compared to household HC (P = 1.3e-05) (Fig. 7*F*). *BA* abundance was not different between the HC and RRMS but was significantly lower in secondary progressive MS (SPMS) and primary progressive MS (PPMS) compared to HC (Fig. 7*G*). Similarly, *AM* abundance was found to be no different between the HC and RRMS but was significantly higher in SPMS and PPMS compared to HC (Fig. 7*H*). Nonetheless, *BA:AM* ratio was significantly lower in PwMS compared to HC, irrespective of the type of MS (Fig. 7*I*). Interestingly, *BA:AM* ratios in SPMS and PPMS were no different from one another and were significantly lower than in patients with RRMS. Additionally, the *BA:AM* ratio in the healthy household was significantly higher irrespective of the treatment status of PwMS (Fig. 7*J*). Strikingly, among PwMS, we observed a significant negative correlation between the *BA:AM* ratio and the EDSS score (R = −0.14, P = 0.0022) (Fig. 7*K*) and disease duration (Fig. 7*M*), further solidifying its potential as a prognostic biomarker. However, no association was observed between *BA:AM* ratio and the Multiple Sclerosis Severity Score (MSSS) (R = −0.083, P = 0.063) (Fig. 7*L*).

While the *BA:AM* ratio also declined with age in both HC (R = −0.14, P = 0.001) and PwMS, (R = −0.23, 1.8e-07), this decline was more pronounced in PwMS (*SI Appendix*, Fig. S9*A*). The *BA:AM* ratio remained significantly lower in PwMS compared to household HC across various demographics and lifestyle factors, including sex, birth method, breastfeeding history, asthma, depression, NSAID use, probiotic use, and pet ownership (*SI Appendix*, Fig. S9 *B* and *I*). On the other hand, the *BA:AM* ratio was significantly lower in PwMS with the following associations or conditions: no history of allergies, no anxiety, no eating disorder, no eczema, no maniac depression disorders, no obsessive-compulsive disorder, no postpartum depression in females, no type 2 diabetes, no other family members with MS, no use of oral contraceptives (OCP_y_n), were former smokers or nonsmoker, and no use of recreational drugs (*SI Appendix*, Fig. S10 *A* and *L*). Although Vitamin D levels had a weak correlation with *BA:AM* ratio, the correlation was lost when the samples were segregated into HC or PwMS groups (*SI Appendix*, Fig. S10 *M* and *N*). These data suggest that the *BA:AM* ratio can be a robust biomarker for PwMS, independent of potential confounding factors in MS.

Discussion

Despite the gut microbiome's known association with MS, there remains limited information on specific microbiota markers for MS. Despite debate on whether gut microbiota causes or results from MS, previous studies have been limited by small sample sizes, low-throughput sequencing technologies, and a lack of validation cohort. Using a comprehensive shotgun metagenomic sequencing analysis in both a human cohort and a mouse model of MS (EAE), we show that the microbiota of PwMS is heterogeneous, with *Blautia* and *Akkermansia* emerging as potential drivers. In contrast, *PC* dominated the microbial community in HC individuals. Utilizing EAE, we identified a lower *Bifidobacterium:Akkermansia* ratio as a key disease biomarker. Furthermore, administering MS-driving *Blautia*

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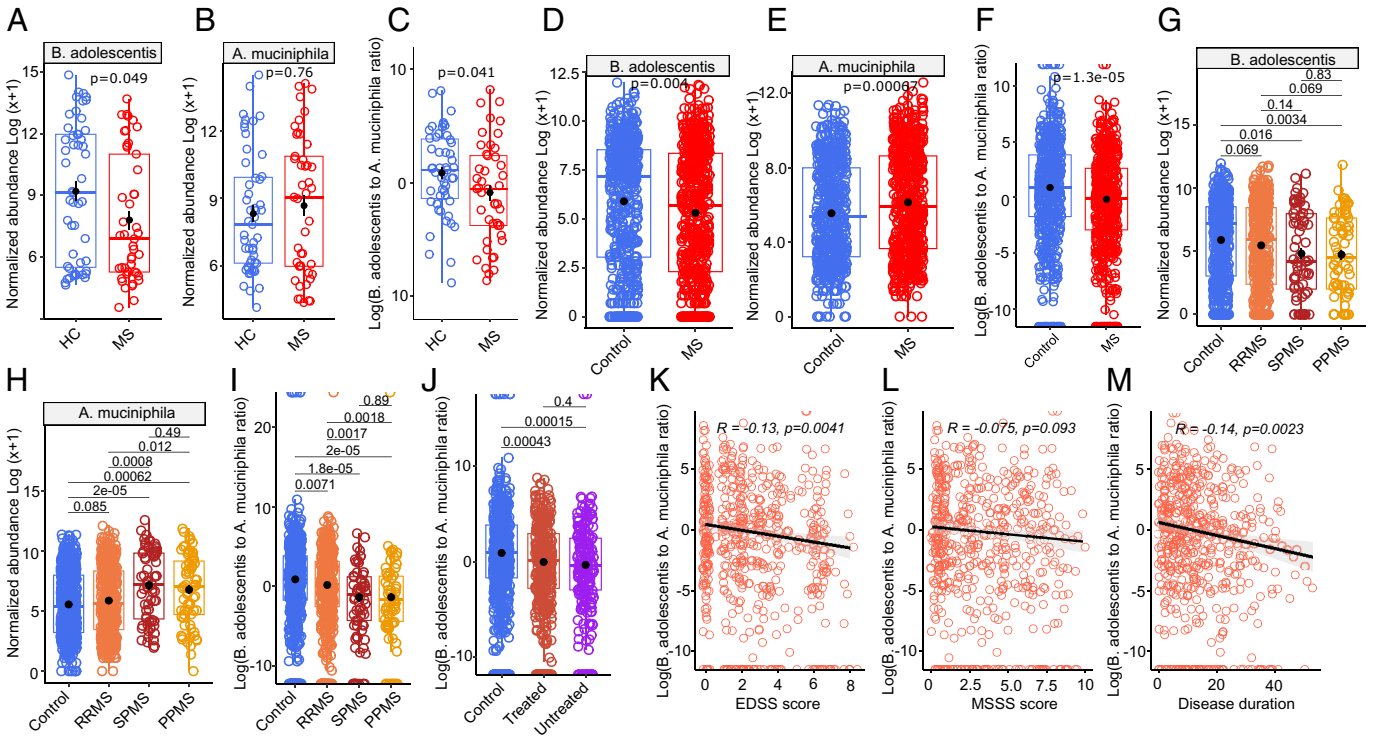


Fig. 7. A lower *B. adolescentis* to *A. muciniphila* ratio (*BA:AM* ratio) is a gut microbial marker in PwMS. Normalized abundance of (*A*) *B. adolescentis* and (*B*) *A. muciniphila*. (*C*) *BA:AM* ratio in HC (n = 51) and PwMS patients (n = 45) from our cohort. Normalized abundance of (*D*) *B. adolescentis* and (*E*) *A.muciniphila*. (*F*) *BA:AM* ratio in the iMSMS cohort (15) consisting of household HCs (n = 576) and PwMS (n = 576). Normalized abundance of (*G*) *B. adolescentis* and (*H*) *A. muciniphila* in the iMSMS cohort (*I*) *BA:AM* ratio in PwMS with different types of the disease, RRMS, SPMS, or PPMS, and (*J*) in healthy household controls and treated and untreated PwMS patients from the iMSMS cohort. Spearman correlation of the *BA:AM* ratio with (*K*) EDSS, (*L*) MSSS, and (*M*) disease duration from the iMSMS cohort. The Kruskal–Wallis test followed by pairwise comparison was performed in (*G–J*), and adjusted *P*-values after "BH" correction are shown.

species into mice reduced *Bifidobacterium:Akkermansia* ratio, induced gut dysbiosis and inflammation, and worsened EAE compared to mice administered HC-driving *PC*. Additionally, the mice experiments established this ratio as a predictor of future disease severity. Importantly, the significance of this ratio extended to PwMS at the species level, where the ratio of *BA:AM* was lower in our PwMS cohort which was further validated in a larger cohort from the iMSMS (15). Collectively, our data highlight the *Bifidobacterium* to *Akkermansia* ratio as a critical gut microbiota feature associated with MS pathogenesis in both PwMS and animal models.

The importance of gut microbiota in MS is supported by evidence that PwMS experience microbial dysbiosis, inflammation, and gastrointestinal symptoms (7, 9, 12, 14–16, 20, 29–31). While specific bacteria could alter disease severity in mice by modulating Th1, Th17 cells, and increasing Tregs (19), the microbial marker that could determine such change is unknown. Our findings align with a recent study highlighting the potential of specific gut bacteria to contribute to MS pathogenesis. Specifically, Ma et al. (32) have shown that PwMS have a higher prevalence of *Clostridium perfringens* strains producing epsilon toxin (ETX), which can contribute to MS pathogenesis by disrupting blood–brain barrier and facilitating immune cell infiltration into the CNS. Our study also demonstrates a similar mechanism, where specific gut bacteria (MS driving *Blautia* species) can alter the gut environment (dysbiotic microbiota and enriched immunostimulatory LPS), leading to increased gut inflammation and disease severity in a mouse model of MS. While we did not specifically investigate ETX, our findings reinforce the concept that gut-derived molecules, whether toxins like ETX or fecal LPS from the dysbiotic community can contribute to MS pathogenesis. Notably, gut microbiome changes, specifically lower *Bifidobacterium* to *Akkermansia* ratio prior to disease onset, strongly suggested that this ratio may be an early indicator of disease severity too. Pre-EAE induction fecal LPS from MS-driving *BW*-administered mice increased proinflammatory cytokine production (e.g., IL6, IL-12/23) in BMDMs, and these elevated levels correlated with later EAE disease severity. Collectively, our findings reveal a strong link between gut dysbiosis and EAE severity, where MS-driving bacteria exacerbated the disease by disrupting the gut microbiome, reducing the crucial *Bifidobacterium* to *Akkermansia* ratio, and enriching for immunostimulatory bacteria that promote proinflammatory cytokine production.

Our data pointed toward a potential pathogenic role for *Blautia* and *Akkermansia* in EAE/MS. Despite potential probiotic properties (33), *Blautia* species have been linked to various inflammatory conditions including colitis, lupus, and irritable bowel syndrome (34–36). While *Blautia* species were reduced in a global iMSMS cohort, lower *Blautia* abundance in *treated* MS patients within the same cohort might suggest a potential benefit associated with loss of *Blautia* (15). Additionally, in a separate MS cohort (Mayo Clinic, MN), we have shown a higher abundance of *Blautia* species in PwMS (9). It is important to consider that both this study and our previous work (9) included MS patients from the Midwestern United States (MN and IA). This geographical restriction may contribute to the observed differences in *Blautia* abundance compared to the iMSMS. Further, Blautia species such as *B. stercoris,* a *Blautia* species has been linked to increased relapse risk in children with MS (37) and *Blautia* administration in mice enriched for potentially proinflammatory *Akkermansia* strains. Both genera were also associated with MS-enriched microbial communities, with *Blautia* assigned to three distinct MS-related topics. These findings underscore the complex role of *Blautia* in MS and the need for further research to clarify its contribution to disease pathogenesis.

Previously, we have shown that an inflammatory diet associated with severe EAE was linked with the enrichment of *Akkermansia*

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and depletion of *Bifidobacterium* (11, 38), highlighting the importance of *Bifidobacterium* to *Akkermansia* ratio in inflammation and a possible pathogenic role for *Akkermansia*. Additionally, *Akkermansia* species such as *A. muciniphila* has been described as a hallmark of MS gut microbiome (8, 12, 15, 31, 39). In contrast, a recent study has suggested that certain strains of *Akkermansia* species can have a protective effect against EAE (10). However, the same study also showed that among all the *Akkermansia* strains, only the H3 strain showed significant reduction in the percentage of RORγt + γδT cells and IL-17 production in mice. The discrepancies between two studies may be attributed to factors such as the lack of data on *Bifidobacterium* in a previous study and the strain-specific effects of *Akkermansia*. Our study emphasizes that the presence of *Bifidobacterium* in the bacterial community appears crucial for maintaining health, and the loss of this species and/or an increase in *Akkermansia*, may contribute to disease. Thus, our findings, in conjunction with the existing literature, highlight the complex and context-dependent role of *Akkermansia* in MS and emphasize the need for further research to fully elucidate its strain-specific effects and interactions with other members of the gut microbiota, such as *Bifidobacterium*. Furthermore, our data on reduced *F. prausnitzii* in PwMS align with previous reports, including the iMSMS cohort (15, 40). Overall, our findings highlight the possibility of multiple microbial species linked to PwMS communities, warranting a deeper investigation of their role in health and disease.

Bifidobacteria make up a major proportion of gut microbiota in adults, where *BA* is a predominant species in >70% of healthy individuals (41, 42). Similar to our cohort, *BA* was previously reported to be higher in HC compared to PwMS (43) or depleted in stool samples from PwMS (44). Further, *Bifidobacterium* was associated with decreased expression of the MS risk allele HLA. DPB1 in controls (45). *BA* can regulate the gut–brain axis through gamma-aminobutyric acid production (46, 47). Findings from human studies have been corroborated in mouse models where the loss of *BA* was linked with increased disease severity in EAE mice (11). In addition, higher abundance of the *Bifidobacterium* in vehicle control compared to strict anaerobe *Faecalibaculum* in the PBS-treated group, highlight its possible role in homeostatic gut inflammatory environment. In contrast, while *AM* is prevalent in healthy individual gut (48), its role in health and disease remains controversial. Multiple studies have shown higher abundance of *AM* in PwMS (8, 10, 12, 30, 31) depicting its negative role. In addition, we have previously shown that a dietary shift from isoflavone-rich foods to a phytoestrogen-free diet in mice resulted in decreased *BA*, an altered *BA:AM* ratio, and worsened EAE severity (11). These findings and our current data suggest opposing roles for *BA* and *AM* within the gut ecosystem. Importantly, both bacteria can strongly adhere to the intestinal mucosal layer (49, 50) to influence immune responses (50–53) and thus may affect MS/EAE severity because of their direct competition for mucosal binding, with their relative abundance impacting disease outcomes. As *Bifidobacterium* strains can influence goblet cell function and improve mucosal barrier acting as a mucin-builder (54), we argue that the disease severity outcome lies in a balance of *AM* and *BA* abundances, especially the rate of mucosal foraging by *Akkermansia* species and mucin building by *Bifidobacterium* species. Higher abundance of mucolytic alpha-L-fucosidase enzyme in the gut microbiota of PwMS in our dataset (*SI Appendix*, Fig. S6*I*) further signifies the importance of the mucosal layer and associated bacteria. However, further research directly involving *AM* and *BA in vitro* and in germ-free mice is warranted to elucidate underlying mechanisms for the regulation pathobiology of MS.

Importantly, the lower *Bifidobacterium* to *Akkermansia* ratio from the mouse study was also extended to PwMS as *BA:AM* ratio was significantly lower in PwMS compared to HCs within our cohort and validated in a separate larger cohort from the iMSMS. The strong negative correlation of the *BA:AM* ratio with EDSS scores and its stability across PwMS with coexisting conditions strengthen its utility as a gut marker in MS, potentially reflecting disease activity independent of multiple health conditions such as allergies, diabetes, psychological disorders, etc.

While our study highlights the *BA:AM* ratio as a predictor of disease severity in mice and correlates with higher EDSS scores and disease duration in a cross-sectional study of iMSMS patients, longitudinal studies are needed to validate its predictive value for disease severity in the broader PwMS population. Additionally, expanding the scope to include other MS types, such as clinically isolated syndrome and progressive MS, as well as longitudinal tracking, would offer a more comprehensive understanding of our findings and their broader utility.

Nonetheless, our study identified significant gut microbial differences between PwMS and HC. We characterized the diverse compositions of the MS gut microbial community and validated the importance of MS-associated and HC-associated bacteria in EAE, an animal model of MS. These findings underscore the importance of the gut–brain axis in MS, suggesting that gut microbial imbalances may contribute to disease development and progression. The *BA:AM* ratio not only offers a promising diagnostic tool but also opens avenues for monitoring disease activity and tailoring treatment strategies for individual patients. By elucidating the functional implications of this microbial signature, our work lays the foundation for developing innovative therapies that target the gut microbiome to potentially alleviate MS symptoms and modify disease course.

Methods

Sex as a Biological Variable. Human fecal samples were obtained from both men and women. However, there were more samples from female patients as MS is three times more prevalent in females than males. The EAE study exclusively examined female mice because male mice develop inconsistent disease. It is unknown whether the findings in female mice are relevant to male mice.

Study Design. This is a matched case and control cross-sectional study where 45 patients with RRMS were recruited from the Neuroimmunology Clinic at the University of Iowa Hospital Clinics from 2015 to 2021. The inclusion criteria for people with MS (PwMS) were a) MS diagnosis using the 2010 revision of McDonald diagnostic criteria and b) 18 to 63 y of age with a diagnosis of relapsing-remitting MS. The exclusion criteria included a) active pregnancy, b) history of bariatric surgery, c) active chronic illness such as inflammatory bowel disease, Celiac disease, etc., and d) use of antibiotics or laxatives. All PwMS were either on DMT or not on treatment for at least 3 mo before enrollment. Age and sex-matched HCs (n = 51) were also enrolled at the University of Iowa. Subjects were asked to provide fecal samples at least 4 wk after their last dose of oral antibiotics or laxatives and at least 3 mo after their last colonoscopy. Demographics and clinical characteristics of all participants providing stool samples at enrollment are provided in Table 1. The study was conducted as per the guidelines approved by the University of Iowa Institutional Review Board. All patients gave informed consent to participate in the study. Additional details regarding the study design are provided in *SI Appendix*.

Quantification of Fecal Biomarkers in Human Fecal Samples. Briefly, 50 mg of fecal sample was mixed in 1 mL of 1% Bovine Serum Albumin, followed by homogenization by vortexing for 2 min and centrifugation at 12,000×g for 2 min to collect fecal supernatant. Plasma was collected from blood samples of HCs and PwMS in the metagenomic study. Fecal lipocalin2 and calprotectin and plasma osteopontin were quantified using commercial ELISA kits (R&D Systems, MN).

Human Fecal Sample Collection and DNA Extraction. Stool samples were collected from HCs and PwMS at their home using Commode Specimen Collection kits (Fisher, PA), shipped overnight on frozen gel packs to the laboratory, and stored at

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−80 °C. Microbial DNA was extracted from each fecal sample using the Qiagen DNeasy PowerLyzer PowerSoil Kit (Qiagen, Germantown, MD) as described previously (11).

Library Preparation, Sequencing, and Quality Control of Human Fecal Metagenomic Reads. Metagenomic shotgun sequencing (2 × 100 bp) was performed using the DNBSEQ-G400 platform (MGI Tech Co., Ltd., Shenzhen, China) at the Bioengineering Lab Co., Ltd., as described previously (55). The obtained paired raw reads were quality controlled using *read_qc* from *metawrap* v1.3.2 (56), aligned to the human genome (*hg38*) to remove host reads with default parameters. The quality-controlled reads were then used for taxonomic and functional analysis. Additional details are provided in *SI Appendix*.

Taxonomy, Taxonomic Topic Modeling, and Functional Assignments of Human Fecal Sequence Reads. The quality-controlled metagenomics reads were assigned to microbial taxonomy using classifier *Kaiju* (57). Reads were searched against the *proGenomes* reference database of protein sequences (downloaded, June 3, 2021) (58). β-diversity was measured using Bray–Curtis dissimilarity metric. For differential analysis, *diff_analysis* function was used from the *microbiotaprocess* package in R with other parameters being default except *mlfun="lda" and lda=3,* $P < 0.05$ and default $q < 0.3$. Both *Bifidobacterium adolescentis* and *Akkermansia muciniphila* were 100% prevalent in all PwMS and HC samples in our cohort and the *BA:AM* ratio at the species level was calculated by dividing the normalized abundance of *BA* to the normalized abundance of *AM*. The ratio was log transformed and plotted.

A topic modeling approach was used on normalized abundance of bacteria to find a suitable number of topics using *FindTopicsNumber* function from *ldatuning* package in R using "*CaoJuan2009*" and "*Arun2010*" metrics and the *VEM* method. A total of 39 topics were identified, which were used for differential abundant testing using the *LDA* function of the *topicmodels* package in R. The top 20 species contributing to differentially abundant topics were visualized using *ggplot2*.

For functional analysis, *humann3* pipeline (59) was used for each sample after concatenating the clean forward and reverse reads using default parameters. Genes were then converted to KEEG Orthologs using *human_regroup_table* function. Differential abundance of the KEGG Orthologs were calculated using the *univariate Wilcoxon test* in R from *microbiomeMarker* package. Differentially expressed KOs were mapped to the gut–brain module (GBM) (60) (download date: 09/03/2023) and Gut Metabolic Module v: 1.07 (GMM) (61) (downloaded 09/03/2023).

As a validation cohort, we obtained sequence taxonomy and metadata (taxa.list.rda) from the shotgun metagenomic dataset of iMSMS consortium study (https://github.com/BaranziniLab/iMSMS_study/tree/main/rdata) (15). Bacterial taxa at the species level not observed at least 100 times in a minimum of 10% of the samples were removed and the samples were normalized using median read counts. *BA:AM* ratio was calculated as described above. Out of 1,152 samples, AM was 99.13% prevalent (10 samples had zero AM reads), while BA was 88.63% prevalent (131 samples had zero reads). Subsequently, the ratio obtained was log(x + 1) transformed and plotted in R using *ggplot2*. For statistical comparisons, "infinite" values were removed. Additional details are provided in *SI Appendix*.

EAE Microbiome Experimental Design, Fecal Shotgun Sequencing, and Downstream Analysis. C57BL/6J female mice (4 to 6 wk old) from Jackson Laboratories (Bar Harbor, ME) were randomly divided into three groups (n = 5 / group) and bedding was mixed for a week to homogenize gut microbiome. EAE was induced as shown in Fig. 3*A* and described previously (62). Additional details are provided in *SI Appendix*.

Fecal samples were collected on day 0, day 7, and day 14, while colonic mucosal samples were collected after the mice were killed on day 14 for microbiome analysis. Microbial taxa were collapsed at the genus level and *lefse* analysis was performed from the *microbiomeMarker* package in R with *lda_cutoff* of 4, *kw_cutoff* of 0.01, and *Wilcoxon_cutoff* of 0.01 for differential abundance of bacteria. To calculate the *Bifidobacterium:Akkermansia* ratio at the genus level, we divided the normalized abundance of *Bifidobacterium* to the normalized abundance of *Akkermansia*. Subsequently, the ratio obtained was log transformed and plotted in R using *ggplot2*.

Experimental Design for *PC*, *PV*, and *BW* Treatment, Fecal 16S rRNA Sequencing, and Analysis. One of the predominant *Blautia* species in the human gut, MS-driving *B. wexlerae* (*BW* DSM19850) (*BW*) (63) and HC-driving *PC* (DSM18205) was acquired from DSMZ (Leibniz Institut DSMZ Germany). *PV* (a bacterium that weighed toward the MS-treated group) was cultured in-house from healthy human fecal samples as described previously (64) and *SI Appendix*.

The microbiome across different groups was homogenized by mixing the bedding for a week. Mice were treated with antibiotics cocktail (combination of 0.5 g/L vancomycin, 1 g/L neomycin, 1 g/L ampicillin, and 1 g/L metronidazole and three sachets of Splenda/L) for a 3-d period to deplete and overcome resistance to colonization of the incoming bacterium as described previously (28). Mice were randomly divided into three groups and gavaged with 0.2 mL (OD_{600} = 0.5) of either *PC, PV*, or *BW* every day for 3 d followed by a single donor FMT from a healthy human fecal sample (0.2 mL). Preprepared healthy human FMT (65) was inoculated to provide diverse microbiome from FMT. Fecal samples from individual mice in all groups were collected on days 9, 13, 17, 22, and 27 followed by EAE induction as shown in Fig. 4*A*. Additional details for *PC*, *PV*, and *BW* treatment are provided in *SI Appendix*.

One hundred forty-five mouse fecal samples were used for DNA extraction and 16S ribosomal RNA (rRNA) (V3-V4 region) amplicon sequencing as described above and previously (66, 67). β-diversity was analyzed using Bray–Curtis dissimilarity metrics and pairwise comparison was made using *pairwiseAdonis* in R. Differential analysis of taxa was performed using *run_test_multiple_groups* function from package *microbioMarker* with *Pvalue_cutoff = 0.05, P_adjust =* "*BH*," and *method =* "*kruskal*" and visualized using *ggplot2* in R. To calculate the *Bifidobacterium:Akkermansia* ratio at the genus level, we divided the normalized compositional abundance of *Bifidobacterium* to the normalized compositional abundance of *Akkermansia*. Subsequently, the ratio obtained was log transformed and plotted in R using *ggplot2*. Detailed method for microbiome analysis is provided in *SI Appendix*.

Extraction of Fecal Lipopolysaccharide from Fecal Samples of *PC*-, *PV*-, or *BW*-Administered Mice. Fecal LPS was isolated from the day 27 fecal samples as described before (11, 68) with slight modification (details in *SI Appendix*).

BMDM Preparation, Lipopolysaccharide Treatment, and Cytokine Profiling. BMDMs were generated as previously described (69) and stimulated with 10 μL of fecal LPS isolated from feces on day 27 of *PC*-, *PV*-, *BW*-administered mice or control (media treatment) for 3 h at 37 °C. Subsequently, cells were washed and cultured with DMEM F12 containing l-glutamine (2 mM), penicillin (100 units/mL), streptomycin (0.1 mg/mL), and incubated at 37 °C for 16 h. Cytokines were measured in the cell-free supernatant by the ELISA.

EAE Induction and Scoring of *PC*-, *PV*-, and *BW*-Administered Mice. EAE induction in *PC*-, *PV*-, or *BW*-administered mice was performed using MOG_{35-55}+CFA+PTX, and clinical symptoms were scored as described in *SI Appendix*.

Statistical Analysis. All analyses were performed using R v4.1.2 and as described above for each section. Specific statistical tests used are described in the legends of respective figures, tables, and datasets.

Data, Materials, and Software Availability. All metagenomic sequences are deposited in NCBI (human shotgun under BioProject PRJNA1084182 (70), EAE shotgun metagenomic sequences under BioProject PRJNA1083036 (71), and 16S rRNA sequences under BioProject PRJNA1086985 (72)). Allother data are included in the manuscript and/or supporting information.

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Competing interest statement: A.K.M. holds a patent (licensed to Evelo Biosciences) on a technology using *Prevotella histicola* for treating autoimmune diseases. No funds/products from this patent were used in this study. Other authors declare no conflict of interest.

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